CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 9: Matrices

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. State the order of $\begin{bmatrix} 4 & 0 & 1 \\ 2 & -3 & 5 \end{bmatrix}$ and the number of its elements.

2. Find a and b if $\begin{bmatrix} a+3 & 4 \\ 6 & 2b \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 6 & 10 \end{bmatrix}$.

3. If $A = \begin{bmatrix} 3 & -2 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 5 \\ 2 & -1 \end{bmatrix}$, find $A - B$.

4. For the matrices in the previous question, find $3A + 2B$.

5. Find the matrix X if $2X + \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 9 & 7 \\ 1 & 3 \end{bmatrix}$.

6. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 4 & 2 \end{bmatrix}$, find AB.

7. For the matrices in the previous question, find BA.

8. If $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, find A^2.

9. Can the product AB be found if A is of order 2×3 and B is of order 2×2? What about BA?

10. Find x and y if $\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \end{bmatrix}$.

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Answer Key

1. 2×3; 6 elements
2. $a = 2, b = 5$
3. $\begin{bmatrix} 3 & -7 \\ -1 & 5 \end{bmatrix}$
4. $\begin{bmatrix} 9 & 4 \\ 7 & 10 \end{bmatrix}$
5. $X = \begin{bmatrix} 4 & 2 \\ -2 & -2 \end{bmatrix}$
6. $\begin{bmatrix} 14 & 4 \\ 1 & -1 \end{bmatrix}$
7. $\begin{bmatrix} 1 & 3 \\ 10 & 12 \end{bmatrix}$
8. $\begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}$
9. AB does not exist; BA exists and is of order 2×3
10. $3x + 2y = 12$ and $x + 4y = 14$ give $x = 2$, $y = 3$

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